

# DOPE: Dynamic Obstacle Parking Environment for End-to-End Autonomous Parking in CARLA

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**Abstract**—End-to-End (E2E) learning has recently attracted growing interest for autonomous parking, as it enables policies to directly map sensor observations to control commands and potentially learn human-like behaviors in complex environments. In particular, such approaches are expected to generalize better than traditional modular pipelines when trained on sufficiently diverse data. However, existing methods are typically trained and evaluated in static, deterministic environments that fail to capture the variability and interactions present in real-world parking scenarios. Although these limitations have been acknowledged, there is currently no standardized simulation framework for E2E parking that systematically incorporates dynamic and stochastic behaviors. To address this gap, we introduce Dynamic Obstacle Parking Environment (DOPE), a closed-loop CARLA-based simulation environment with dynamic scenarios designed to evaluate E2E parking models under interactive conditions. Rather than inducing failures through adversarial or overly aggressive behaviors, DOPE introduces controlled stochasticity and state-based dynamic interactions that reflect common perturbations in a parking lot. Using this framework, we evaluate several state-of-the-art E2E parking models that were not trained under such complex conditions. Our results show that model performance degrades noticeably even with the introduction of stochasticity alone, and deteriorates further under dynamic interactions with surrounding vehicles. These results indicate that current E2E parking models are highly sensitive to distribution shift and struggle to generalize to various environments. Our findings highlight the limitations of existing evaluation practices and demonstrate that DOPE provides a practical step toward more robust and generalizable autonomous parking systems. The code will be made available upon the acceptance.

## I. INTRODUCTION

With around 20% of vehicle crashes taking place in parking environments such as lots and garages [1], [2], advancing autonomous parking technologies could offer opportunity to enhance traffic safety. Recent works on E2E learning-based autonomous parking [3], [4], [5], [6] demonstrates that such models can learn human-like parking maneuvers in simulation, which indicates their potential to improve the safety and reliability of autonomous parking.

Training and evaluating such models in real-world environments is both costly and potentially unsafe, and therefore simulation becomes a practical and scalable alternative [7]. In particular, closed-loop simulation, where the model interacts with the environment and its actions influence future states, is essential for assessing control performance and safety in

sequential decision-making tasks such as parking. However, existing simulation-based evaluations for parking are typically conducted in static and deterministic environments that lack interactions and variability. For example, parked vehicles are often perfectly aligned, and surrounding traffic is either absent or rarely perturbs the Ego Vehicle (EV). While such setups are sufficient for evaluating geometric accuracy and control execution, they provide limited insight into how models behave under interactive conditions.

In real parking environments, vehicles frequently interact in narrow aisles, and parked cars are often misaligned within their slots. These factors introduce uncertainty and require adaptive decision-making, which are largely absent in simplified simulation settings. Since most E2E parking models are trained using Imitation Learning (IL) and/or Reinforcement Learning (RL), the diversity and realism of training and evaluation environments play a critical role in determining policy robustness. Consequently, models that perform well in static benchmarks may struggle when exposed to dynamic objects and stochastic events, potentially leading to unsafe or unpredictable behaviors [6], [8], [9].

To address this limitation, we introduce DOPE, a dynamic parking scenario suite in CARLA [10] designed to evaluate autonomous parking policies under interactive conditions. To our knowledge, this is the first framework that systematically studies the impact of stochastic and interactive dynamics in E2E parking. Rather than deliberately inducing failures through adversarial perturbations [11], our objective is to expose EVs to complex situations commonly encountered in real parking lots for E2E models to handle safety-critical interactions and ultimately improve their robustness. DOPE comprises an optional misaligned vehicle setting and three dynamic scenarios that are both common and challenging for decision-making: (i) adjacent vehicle departure (a parked vehicle suddenly exits its slot), (ii) on-path following (a trailing vehicle moves within a shared aisle with stochastic stop-and-go behavior), and (iii) transient lane obstruction (a vehicle temporarily blocks access to the target parking slot). These scenarios are designed to capture typical parking lot interactions that require drivers to reason about whether to yield, wait, or proceed under uncertainty, often in conditions of limited visibility and tight spatial constraints. To this end, the proposed scenarios incorporate stochastic and dynamic obstacles that emulate these interactions, as well as variations in parked vehicle alignment that reflect imperfect real-world parking behavior. At the same time, the scenarios remain fully controllable and reproducible to support systematic evaluation. Moreover, DOPE is implemented within a uni-

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fied framework that enables existing E2E parking models to be integrated with minimal effort.

Using DOPE, we reevaluate representative E2E parking models under stochastic and dynamic conditions and compare their performance against conventional static benchmarks. Our experiments reveal that even the introduction of stochasticity alone, such as randomized ego initialization and misaligned parked vehicles, leads to noticeable performance degradation. When dynamic obstacles are further introduced, performance deteriorates significantly, and the collision rate increases. Our findings demonstrate that current E2E parking models are highly sensitive to distribution shifts and fail to generalize under various conditions. This exposes the limitations of existing evaluation practices and underscores the necessity of training and testing these models in stochastic, interactive environments. Consequently, this work provides critical guidance for future research in E2E parking, emphasizing the need to integrate dynamic interactions and stochastic variability into both training and evaluation pipelines.

The contribution of this paper is as follows:

- We introduce DOPE, a CARLA-based dynamic parking scenario suite that models stochastic and interactive behaviors commonly encountered in real-world parking environments, including misaligned parking layouts and dynamic vehicle interactions.
- We provide a unified and easily extensible evaluation framework that enables seamless integration of existing E2E parking models for closed-loop testing under safety-critical conditions.
- We conduct a systematic evaluation of representative state-of-the-art E2E parking models under stochastic and dynamic scenarios, without retraining.
- We demonstrate that current E2E parking models are highly sensitive to distribution shift, with significant performance degradation under stochastic and interactive conditions that are not observable in static benchmarks.

## II. RELATED WORKS

CARLA [10] is one of the most widely used open-source simulators for autonomous driving research that offers realistic physics, diverse sensor suites, and configurable HD-map-based towns with parking lots. It supports standardized benchmarks, such as leaderboards for route navigation and interactive scenarios, enabling scalable policy training and evaluation under controlled conditions [6].

CARLA Real Traffic Scenarios (CRTS) [12] extends CARLA with a catalog of interactive traffic scenarios, such as highway lane changes and roundabouts, designed to stress-test tactical driving behaviors like merging and right-of-way negotiation. CARLA’s built-in Traffic Manager [13] orchestrates background vehicles in autopilot mode to generate dynamic traffic conditions, using rule-based behavior with configurable parameters for traffic flow and rule compliance.

Beyond task-specific benchmarks, there is growing interest in generic scenario generation frameworks for CARLA. ScenarioRunner [14], CARLA’s official scenario definition and execution engine, supports scripted traffic sce-

narios via Python or OpenSCENARIO XML format [15]. Scenic [16] provides a probabilistic domain-specific language for creating and configuring physical objects and agents through abstract, high-level specifications. Complementary to these code-centric approaches, an interactive no-code framework [17] provides a graphical interface for both manual and automated creation of traffic scenarios by representing OpenDRIVE maps [18] as graphs and supporting random sampling of actor types, behaviors, and environmental conditions.

Nevertheless, these tools [12], [13], [14], [16], [17] are primarily designed for structured on-road navigation scenarios and provide limited support for fine-grained maneuvers in unstructured parking environments. Unlike road traffic, parking environments often lack explicit traffic control signals such as traffic lights and stop signs, and vehicle behavior is less structured and can be more unpredictable. Existing tools are optimized for structured traffic behaviors such as lane following, junction negotiation, and leading-vehicle interactions, whereas parking scenarios demand finer spatial control and interaction modeling at the level of individual parking slots, including slot occupancy, goal-specific evaluation criteria, reversing maneuvers, and tight spatial interactions within confined geometries. As a result, adapting these frameworks to parking requires substantial additional modeling effort beyond their original design assumptions.

In contrast, [19] studies parking as a multi-agent simulation problem aimed at improving parking efficiency and evaluating parking and unparking trajectories. However, this work is built on the Dragon Lake Parking (DLP) [20] dataset rather than CARLA, and therefore does not provide sensor-level simulation or a CARLA-compatible environment for perception-based evaluation. Moreover, it focuses on planning and trajectory execution from existing vehicle states, rather than providing a full simulation stack for E2E parking policy training and sensor-driven interaction modeling.

SEG-Parking [6] advances interactive parking simulation in CARLA by introducing two dynamic scenarios in a narrow aisle. One scenario places a low-priority Non-Player Character (NPC) that yields to the EV, which requires the EV to confidently initiate parking. Another scenario features a high-priority NPC that proceeds first, forcing the EV to recognize priority and wait patiently. However, the scenarios remain narrowly scoped and lack diverse dynamics, such as active perturbations during parking execution.

These limitations motivate the need for dedicated scenario frameworks that explicitly model interactive and stochastic dynamics in parking environments. To our knowledge, such systematic evaluation has not been previously explored for E2E parking. To this end, we introduce DOPE, a dynamic parking scenario suite based on CARLA, designed to go beyond generic traffic generation by explicitly modeling complex interactions commonly encountered in parking environments. The framework provides a set of parking scenarios with controlled stochasticity in surrounding vehicles, enabling systematic evaluation of E2E parking policies under interactive conditions. While the current scenarios are

implemented using state machines with a single dynamic object, they represent an initial step toward more general and automated generation of interactive parking environments.

### III. DYNAMIC PARKING SCENARIOS

DOPE is built on the same open-source autonomous parking codebase used by prior E2E parking works [3], [4], [5] for direct comparison with existing models and easier integration of future E2E approaches. As shown in Figure 1, the environment is based on the public parking lot map in Town04\_Opt. Candidate target parking slots are located in rows 2 and 3, where odd-numbered slots are reserved for validation and even-numbered slots are used for training. Background NPC vehicles are spawned at randomly sampled parking slots, with the number of vehicles randomly selected such that at least one third of the slots are occupied. The target slot is excluded from this sampling process. The EV is initialized at randomly sampled spawn positions near the target slot, with its heading randomly sampled but constrained to remain perpendicular to the parking slots.

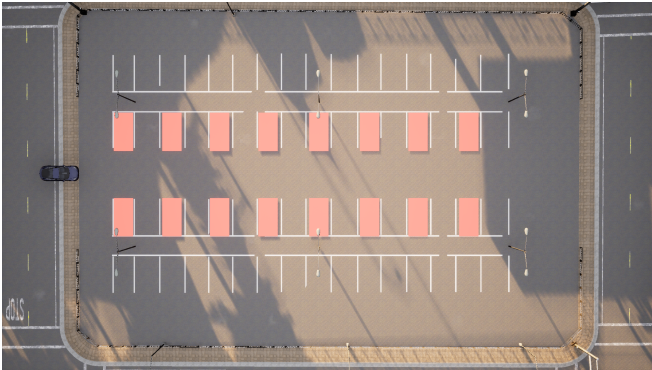


Fig. 1: Layout of Town04\_Opt and available target parking slots for evaluation

To ensure reproducibility while maintaining controlled stochasticity, we employ a deterministic seeding for all sources of randomness in the simulation. Specifically, a global base seed is defined for each experiment, and individual random components (e.g., EV spawn position, NPC placement, and behavior parameters) are generated using the base seed combined with fixed offsets. This design guarantees that each experimental run is fully reproducible, while still enabling diverse and systematically varied configurations across trials.

Based on this setup, DOPE introduces a set of configurable scenario modes designed to model diverse and safety-critical interactions in parking environments. These include an optional misaligned vehicle mode that increases spatial variability, as well as three dynamic NPC behaviors that capture common parking lot interactions: adjacent vehicle departure, on-path following, and transient lane obstruction.

#### A. Misaligned Vehicles

To better reflect real-world variability, DOPE includes an optional misaligned vehicles mode in which background

vehicles are not perfectly aligned within their parking slots. When enabled, parked NPC vehicles are initialized with small pose perturbations (i.e., position and orientation offsets), as shown in Figure 2. This option transforms the map from an idealized layout into a more natural parking lot configuration, which may increase the difficulty of parking without colliding with adjacent vehicles. In addition, the misaligned vehicle setting may reduce the number of parked vehicles due to increased spatial constraints.

It is to be noted that these perturbations do not affect the evaluation, as any resulting offset in the final parking position remains within the defined success criteria. Furthermore, the perturbations are bounded so that any overlap does not prevent a successful parking maneuver.



(a) Perfectly parked vehicles

(b) Misaligned vehicles

Fig. 2: Comparison between perfectly aligned and misaligned scenarios

#### B. Adjacent vehicle departure (Drive-Out)

In the *drive-out* scenario, DOPE models a safety-critical interaction in which a nearby parked vehicle begins to leave its slot while the EV is approaching or preparing to park. At least one vehicle is spawned adjacent to the target slot, and for each target slot, either the left or right neighboring vehicle (if present) is randomly selected to depart. As illustrated in Figure 3a, the departing NPC executes a staged maneuver: it first moves forward out of the slot, then turns toward a randomly sampled direction (i.e., left or right) and stops after completing the turn. This randomness in turn direction may cause the NPC to either temporarily block the EV or leave the path unobstructed. This scenario captures a common parking lot hazard in which a vehicle suddenly exits its slot, altering the available free space and requiring the EV policy to react safely and adapt its trajectory online. In particular, the EV should optimally infer whether the departing NPC constitutes an active obstruction and decide whether to yield or proceed accordingly.

In the current implementation, all NPCs are reverse-parked to simplify the drive-out behavior. Future work will extend this scenario to include vehicles that are forward-parked.

#### C. On-path following (Follow)

In *follow* mode, DOPE introduces a trailing vehicle that shares the driving lane with the EV throughout the parking maneuver. At the start of each run, the NPC is spawned either ahead of or behind the EV (on the same lateral axis), as illustrated in Figure 3b. The spawn position of the NPC

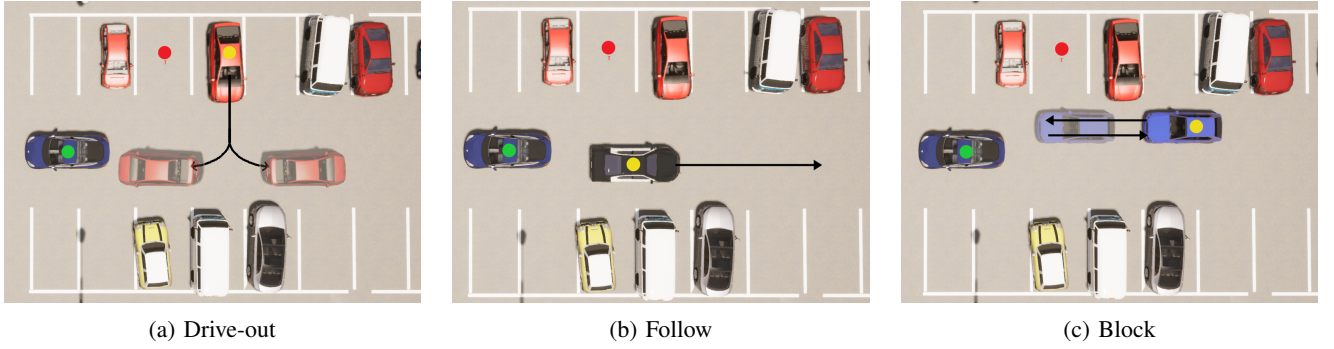


Fig. 3: Three dynamic NPC behaviors in DOPE (The EV is shown in green, the dynamic NPC in yellow, and the target parking slot in red)

is conditioned on the EV’s initial position and direction to ensure that it lies along the EV’s intended path while avoiding unintended blockage of the EV’s maneuver.

Since the evaluated models perform reverse parking, the EV may either directly reverse into the slot or first move forward before reversing, and therefore spawning the NPC accordingly maximizes interaction during the maneuver. The NPC then moves along the lane using a stochastic stop-and-go policy, alternating between acceleration and braking phases of randomized duration to produce irregular following behavior. This models a common real-world scenario in which a nearby vehicle moves unpredictably within a shared parking-lot lane.

#### D. Transient lane obstruction (Block)

In *block* mode, DOPE deploys a NPC that intentionally interferes with access to the target slot corridor. At the start of each run, a candidate blocker is spawned near the target lane with a randomized longitudinal offset and an orientation selected to cross the slot entrance. The NPC then drives into the corridor and brakes to create a temporary blockage. After a randomized blocking timeout, it transitions to a clearing maneuver: it moves away from the corridor, choosing reverse or forward motion based on the EV’s relative position (reverse if the EV is in front, forward if the EV is behind). This creates a transient bottleneck at the slot entrance, requiring the EV to wait, yield, or adapt its approach timing.

In drive-out and block modes, the behavior of dynamic NPCs is conditioned on their proximity to the EV, incorporating safety-based braking when the distance becomes too small. This design encourages the EV to adapt to the current situation and safely complete the parking maneuver, rather than having NPCs induce failures through aggressive or unavoidable collisions. However, these safety constraints may occasionally limit the intended dynamic obstruction. For example, in the drive-out mode, the departing NPC may remain stationary if the EV is spawned in front of it and blocks its exit path.

## IV. BENCHMARKING

To assess the importance of dynamic scenarios for E2E parking, we evaluate three publicly available IL-based E2E parking models: E2E Parking [3], CAA Policy [4], and Dino-Diffusion Parking (DDP) [5]. We exclude the RL-based SEG Parking [6], as its implementation is not publicly available. It is worth noting that E2E Parking uses four cameras, whereas CAA Policy and DDP use six cameras. While DOPE is compatible with other CARLA versions, we use version 0.9.11 to maintain consistency and ensure fair comparison with prior work. All models are evaluated using checkpoints with 4 epochs over 16 target parking slots, with 6 trials per slot under different EV spawn positions and orientations.

To provide additional context, we include both the performance metrics reported in the original papers and the reproduced results under a pseudo-deterministic setting consistent with prior evaluation protocols. These results are obtained in static environments with deterministic ego initialization, enabling a more direct comparison with previously reported performance.

The reported performance of E2E Parking does not correspond to the exact checkpoint used in our evaluation due to differences in the released models and data availability. The original paper reports higher performance, whereas the publicly available checkpoint exhibits lower performance. Therefore, the reported results should be interpreted as a reference rather than a direct comparison. For DDP, we use the publicly available checkpoint corresponding to the DDP-emb+seg configuration.

As a baseline, we evaluate the models in a static environment without misparking or dynamic interactions, while introducing randomized ego initialization. Specifically, the EV spawn position and heading are sampled within the same predefined range used in prior work, replacing the deterministic setup. This allows us to assess the models’ generalization under mild stochasticity while keeping the environment otherwise unchanged.

We then evaluate all models without retraining under four complex settings with stochasticity and dynamic interactions. Specifically, we consider: (i) misaligned vehicles only (Misaligned), (ii) misaligned vehicles with drive-out interactions



TABLE I: Performance comparison of E2E parking models under different scenarios.

| Model           | Setting                 | TSR (%) $\uparrow$ | NTSR (%) $\downarrow$ | CR (%) $\downarrow$ | TR (%) $\downarrow$ |
|-----------------|-------------------------|--------------------|-----------------------|---------------------|---------------------|
| E2E Parking [3] | Reported <sup>a</sup>   | 91.41              | 3.39                  | 2.08                | 1.04                |
|                 | Reproduced <sup>b</sup> | 82.55 $\pm$ 4.16   | 4.17 $\pm$ 1.94       | 7.29 $\pm$ 2.65     | 2.86 $\pm$ 1.36     |
|                 | Baseline <sup>c</sup>   | 72.14 $\pm$ 4.23   | 3.65 $\pm$ 1.74       | 14.58 $\pm$ 3.46    | 2.86 $\pm$ 1.22     |
|                 | Misaligned              | 61.46 $\pm$ 4.42   | 4.17 $\pm$ 1.26       | 26.30 $\pm$ 4.01    | 3.65 $\pm$ 1.54     |
|                 | Misaligned + Drive-Out  | 16.67 $\pm$ 3.42   | 4.43 $\pm$ 0.77       | 71.88 $\pm$ 4.59    | 5.73 $\pm$ 1.81     |
|                 | Misaligned + Follow     | 10.42 $\pm$ 1.93   | 3.12 $\pm$ 0.74       | 78.12 $\pm$ 3.54    | 6.51 $\pm$ 2.01     |
|                 | Misaligned + Block      | 9.64 $\pm$ 3.02    | 5.21 $\pm$ 1.92       | 79.95 $\pm$ 4.50    | 4.17 $\pm$ 2.07     |
| CAA Policy [4]  | Reported <sup>a</sup>   | 87.5 $\pm$ 3.7     | 0.8 $\pm$ 0.6         | 3.5 $\pm$ 2.6       | 3.8 $\pm$ 2.2       |
|                 | Reproduced <sup>b</sup> | 87.24 $\pm$ 4.12   | 0.00 $\pm$ 0.00       | 5.21 $\pm$ 2.00     | 4.95 $\pm$ 2.55     |
|                 | Baseline <sup>c</sup>   | 71.35 $\pm$ 3.56   | 1.56 $\pm$ 0.37       | 22.92 $\pm$ 2.26    | 2.86 $\pm$ 1.51     |
|                 | Misaligned              | 68.75 $\pm$ 3.53   | 1.82 $\pm$ 0.93       | 22.14 $\pm$ 2.55    | 3.91 $\pm$ 2.19     |
|                 | Misaligned + Drive-Out  | 43.23 $\pm$ 3.52   | 0.26 $\pm$ 0.18       | 54.17 $\pm$ 3.03    | 1.30 $\pm$ 0.92     |
|                 | Misaligned + Follow     | 28.65 $\pm$ 3.00   | 1.56 $\pm$ 0.37       | 67.45 $\pm$ 2.72    | 0.78 $\pm$ 0.55     |
|                 | Misaligned + Block      | 19.27 $\pm$ 1.47   | 0.78 $\pm$ 0.18       | 76.56 $\pm$ 1.32    | 1.30 $\pm$ 0.77     |
| DDP [5]         | Reported <sup>a</sup>   | 92 $\pm$ 11        | 0 $\pm$ 0             | 0 $\pm$ 0           | 4 $\pm$ 10          |
|                 | Reproduced <sup>b</sup> | 92.71 $\pm$ 1.32   | 0.00 $\pm$ 0.00       | 0.00 $\pm$ 0.00     | 3.91 $\pm$ 0.92     |
|                 | Baseline <sup>c</sup>   | 84.38 $\pm$ 1.27   | 0.00 $\pm$ 0.00       | 6.25 $\pm$ 0.00     | 5.99 $\pm$ 1.43     |
|                 | Misaligned              | 80.47 $\pm$ 1.13   | 0.00 $\pm$ 0.00       | 6.77 $\pm$ 0.21     | 6.25 $\pm$ 1.27     |
|                 | Misaligned + Drive-Out  | 57.03 $\pm$ 1.03   | 0.00 $\pm$ 0.00       | 38.02 $\pm$ 0.21    | 2.08 $\pm$ 0.58     |
|                 | Misaligned + Follow     | 41.15 $\pm$ 1.07   | 0.00 $\pm$ 0.00       | 56.25 $\pm$ 0.37    | 0.26 $\pm$ 0.18     |
|                 | Misaligned + Block      | 24.48 $\pm$ 0.95   | 0.00 $\pm$ 0.00       | 72.66 $\pm$ 0.77    | 0.78 $\pm$ 0.18     |

<sup>a</sup> Reported results correspond to the original evaluation reported in prior work, which use pseudo-deterministic ego initialization and static environments.

<sup>b</sup> Reproduced results are obtained using the publicly available checkpoints under the same pseudo-deterministic setup for fair comparison.

<sup>c</sup> Baseline refers to a static environment without dynamic interactions or misaligned vehicles, where the ego vehicle is initialized at randomly sampled positions and orientations near the target parking slot.

<sup>c</sup> DDP-emb+seg

TSR: Target Success Rate; NTSR: Non-Target Success Rate; CR: Collision Rate; TR: Timeout Rate.

$\uparrow$  indicates higher is better;  $\downarrow$  indicates lower is better.

(Misaligned + Drive-Out), (iii) misaligned vehicles with follow behavior (Misaligned + Follow), and (iv) misaligned vehicles with transient blocking (Misaligned + Block). We include misaligned vehicles in all settings, as perfectly aligned parking configurations are unrealistic in real-world environments.

We evaluate model performance using four metrics commonly adopted in prior work:

- Target Success Rate (TSR) measures the fraction of trials in which the EV successfully parks in the designated target slot. A trial is considered successful if the distance between the center of the EV and the center of the target slot is within 0.6m laterally and 1.0m longitudinally, and the orientation error is less than  $10^\circ$ .
- Non-target Success Rate (NTSR) measures the fraction of trials that the EV parks in an unoccupied slot that is not the designated target.
- Collision Rate (CR) measures the fraction of trials that any collision occurs during the parking maneuver.
- Timeout Rate (TR) measures the fraction of trials that the EV fails to complete the parking task within a predefined time horizon of 90 consecutive frames.

As shown in Table I, we observe a consistent performance gap between the results reported in prior work, our reproduced results, and the proposed baseline. While the reproduced results generally follow the trends reported in the original papers under pseudo-deterministic settings, a noticeable performance drop is observed when moving to

our baseline setting. Specifically, all models exhibit lower TSR and higher CR under the baseline compared to both reported and reproduced results.

This degradation indicates that even mild stochasticity introduced through randomized ego initialization leads to a distribution shift from the training conditions, resulting in reduced performance. The gap between reproduced and baseline results suggests that existing E2E parking models have limited generalization beyond pseudo-deterministic setups, even when the environment remains static.

Comparing the baseline with the misaligned setting, we observe a further decline in performance, characterized by decreased TSR and increased CR. This highlights the impact of spatial perturbations in parked vehicles, where misalignment introduces tighter constraints and increases the likelihood of collision.

The introduction of dynamic NPCs leads to a significant additional drop in performance across all models. As interaction complexity increases, TSR decreases sharply while CR rises, indicating that current models struggle to handle dynamic and interactive environments. In particular, scenarios with stronger obstruction, such as transient blocking of the target slot, result in the most severe degradation. For example, TSR drops to 9.64% for E2E Parking, 19.27% for CAA Policy, and 24.48% for DDP.

These performance degradations can be attributed to covariate shift [8], where the state distribution encountered during deployment differs from the distribution observed

during training. In imitation learning, this mismatch can cause compounding errors, since policies may encounter out-of-distribution states that were rarely or never seen during demonstrations, especially when the deployment environment contains stochasticity or dynamic interactions.

Among the evaluated models, DDP consistently achieves the highest performance under DOPE, maintaining relatively higher TSR and exhibiting zero NTSR across all scenarios, indicating more precise target selection. However, even DDP experiences substantial degradation under dynamic conditions, underscoring the overall challenge of robust E2E parking in interactive environments.

These results reveal that current E2E parking models rely heavily on deterministic assumptions and fail to generalize to various environments. This highlights the need for incorporating dynamic and stochastic scenarios during both training and evaluation.

## V. CONCLUSION

In this work, we present DOPE, a closed-loop simulation environment with dynamic scenarios built on CARLA for evaluating E2E autonomous parking models under interactive conditions. Unlike existing benchmarks that rely on static and deterministic setups, DOPE introduces controlled stochasticity and dynamic interactions that better reflect real-world parking environments while maintaining reproducibility. By evaluating state-of-the-art E2E parking models, we demonstrate that performance degrades significantly when transitioning from static to stochastic and dynamic scenarios, highlighting the limitations of current approaches and the importance of various evaluation settings.

DOPE provides a practical step toward bridging the gap between simplified simulation environments and real-world parking complexity. As an easily integrable simulation framework with diverse interactive conditions, it offers a useful tool for developing more robust and generalizable parking policies.

In future work, we plan to extend DOPE along several directions. First, we aim to support a wider range of maps to increase environmental diversity. Second, we plan to design additional dynamic scenarios that further challenge E2E parking models under more complex interactions that incorporates multiple dynamic vehicles. Finally, we will explore agent-based control for NPCs, replacing the current state-machine-driven behaviors to enable more realistic and adaptive interactions.

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